

Q2 Customer Feedback System 50 marks

Design an Applet-based GUI to collect customer feedback using input fields, rating options, and submission buttons. Implement event handling to process user inputs and dynamically update the display of feedback results. Ensure proper validation of inputs before submission. Analyze the limitations of Applet technology in modern applications, including security and browser compatibility issues. Evaluate the effectiveness of the chosen UI design and layout managers. Suggest modern alternatives such as web-based or desktop frameworks for improved performance and scalability.

Code:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class CustomerFeedback extends JFrame implements ActionListener {

    JTextField nameField;
    JComboBox<String> ratingBox;
    JTextArea feedbackArea;
    JButton submitButton;
    JLabel resultLabel;

    public CustomerFeedback() {

        // Window settings
        setTitle("Customer Feedback System");
        setSize(600, 450);
        setLocationRelativeTo(null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // Main panel
        JPanel mainPanel = new JPanel(new BorderLayout(15, 15));
        mainPanel.setBorder(BorderFactory.createEmptyBorder(20, 25, 20, 25));

        // Title
        JLabel title = new JLabel(
            "Customer Feedback System",
            SwingConstants.CENTER
        );

        title.setFont(new Font("Arial", Font.BOLD, 24));
        mainPanel.add(title, BorderLayout.NORTH);
```

```
// Form panel
JPanel formPanel = new JPanel(new GridBagLayout());

GridBagConstraints gbc = new GridBagConstraints();
gbc.insets = new Insets(8, 8, 8, 8);
gbc.fill = GridBagConstraints.HORIZONTAL;

// Customer Name
JLabel nameLabel = new JLabel("Customer Name:");
nameLabel.setFont(new Font("Arial", Font.BOLD, 14));

nameField = new JTextField(20);

gbc.gridx = 0;
gbc.gridy = 0;
gbc.weightx = 0;
formPanel.add(nameLabel, gbc);

gbc.gridx = 1;
gbc.weightx = 1;
formPanel.add(nameField, gbc);

// Rating
JLabel ratingLabel = new JLabel("Rating:");
ratingLabel.setFont(new Font("Arial", Font.BOLD, 14));

String[] ratings = {
    "Excellent",
    "Good",
    "Average",
    "Poor"
};

ratingBox = new JComboBox<>(ratings);

gbc.gridx = 0;
gbc.gridy = 1;
gbc.weightx = 0;
formPanel.add(ratingLabel, gbc);

gbc.gridx = 1;
gbc.weightx = 1;
formPanel.add(ratingBox, gbc);
```

```

// Feedback
JLabel feedbackLabel = new JLabel("Feedback:");
feedbackLabel.setFont(new Font("Arial", Font.BOLD, 14));

feedbackArea = new JTextArea(6, 20);
feedbackArea.setLineWrap(true);
feedbackArea.setWrapStyleWord(true);

JScrollPane scrollPane = new JScrollPane(feedbackArea);

gbc.gridx = 0;
gbc.gridy = 2;
gbc.weightx = 0;
gbc.weighty = 1;
gbc.anchor = GridBagConstraints.NORTH;
formPanel.add(feedbackLabel, gbc);

gbc.gridx = 1;
gbc.weightx = 1;
gbc.fill = GridBagConstraints.BOTH;
formPanel.add(scrollPane, gbc);

mainPanel.add(formPanel, BorderLayout.CENTER);

// Bottom panel
JPanel bottomPanel = new JPanel(new BorderLayout(10, 10));

submitButton = new JButton("Submit Feedback");
submitButton.setFont(new Font("Arial", Font.BOLD, 14));
submitButton.addActionListener(this);

resultLabel = new JLabel(
    " ",
    SwingConstants.CENTER
);

resultLabel.setFont(new Font("Arial", Font.BOLD, 13));

bottomPanel.add(submitButton, BorderLayout.NORTH);
bottomPanel.add(resultLabel, BorderLayout.SOUTH);

mainPanel.add(bottomPanel, BorderLayout.SOUTH);

add(mainPanel);

```

```

        setVisible(true);
    }

    @Override
    public void actionPerformed(ActionEvent e) {

        String name = nameField.getText().trim();
        String feedback = feedbackArea.getText().trim();
        String rating = (String) ratingBox.getSelectedItem();

        // Validation
        if (name.isEmpty()) {
            resultLabel.setText("Please enter customer name.");
            return;
        }

        if (feedback.isEmpty()) {
            resultLabel.setText("Please enter feedback.");
            return;
        }

        // Display result
        resultLabel.setText(
            "Thank you " + name +
            "! Rating: " + rating +
            " | Feedback submitted successfully."
        );
    }

    public static void main(String[] args) {

        SwingUtilities.invokeLater(() -> {
            new CustomerFeedback();
        });
    }
}

```

Customer Feedback System

Customer Name :

Swarna

Rating:

Excellent

Feedback:

Very good service. I am satisfied with the product.

Submit Feedback

Thank you Swarna! Rating: Excellent | Feedback submitted successfully.

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Mostly cloudy

Search



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